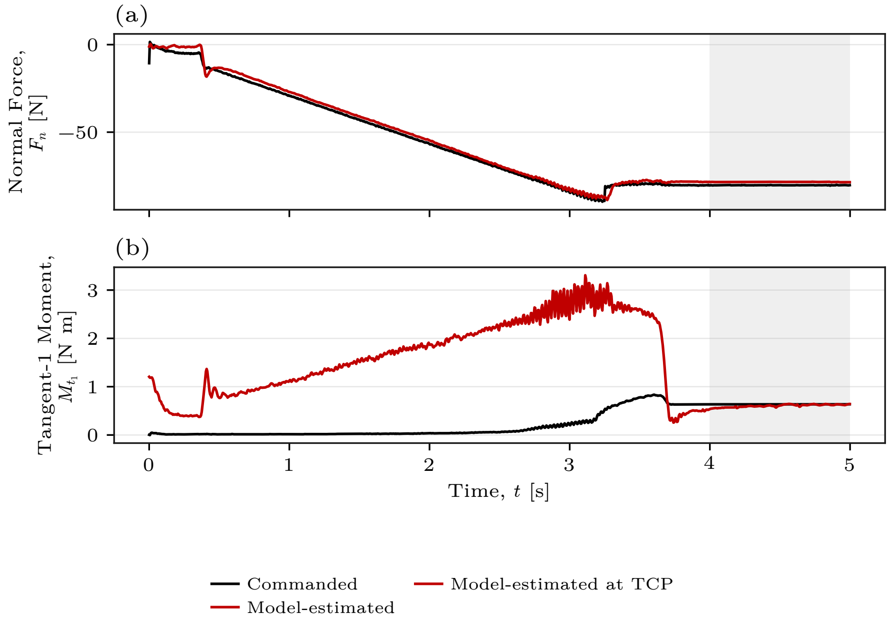
Contact-Establishment Wrench Trace, $r_c = 0$

Used impedance. The following gains were applied in the surface basis $[t_1, t_2, n_s]$:

$$K_p = [2000,\ 2000,\ 350]\,\mathrm{N/m}, \qquad D_p = [231.2,\ 177.9,\ 112.1]\,\mathrm{N\,s/m},$$
$$K_R = [5,\ 5,\ 50]\,\mathrm{N\,m/rad}, \qquad D_R = [1.5,\ 1.9,\ 1.2]\,\mathrm{N\,m\,s/rad}.$$

The damping values were calculated automatically at Contact Establishment entry with damping factor 1.9; the manual damping values were not used as lower bounds. This contact sequence used null-space mode 3. The centre of compliance was placed at the TCP, with $r_c = p_c - p_{\mathrm{TCP}} = 0$. The first tangent t_1 is the normalised projection of the base-frame x_0-direction onto the configured surface plane:

$$t_1 = \frac{e_{x_0} - (n_s \cdot e_{x_0}) \cdot n_s}{\|e_{x_0} - (n_s \cdot e_{x_0}) \cdot n_s\|} = (0.999851, -0.000477, -0.017238)^\top.$$



The curves show one five-second Contact Establishment trace. The grey area from 4 to 5 s is the stationary interval used for the measured means. Both plotted signals are absolute quantities resolved on the same surface axes. The physical contact force is denoted throughout by ${}^{O}f_{\mathrm{contact}}$.

Normal-force comparison, F_n

The mean desired and measured TCP positions in the grey interval were

$$p_d = (0.511749, -0.005127, -0.210071)^\top\,\mathrm{m},$$
$$p_{\mathrm{TCP}} = (0.513909, 0.005024, 0.019199)^\top\,\mathrm{m}.$$

Their normal error gives

$$e_n = n_s^\top \cdot (p_d - p_{\mathrm{TCP}}) = -0.229466\,\mathrm{m},$$
$$F_{n,\mathrm{qs}} = K_{p,n} \cdot e_n = 350 \cdot (-0.229466) = -80.313\,\mathrm{N},$$
$$F_{n,\mathrm{cmd}} = -80.312\,\mathrm{N}, \qquad F_{n,\mathrm{est}} = -78.451\,\mathrm{N}.$$

The estimate differs from the command by 1.861 N, or 2.32%. The quasi-static value and the full command differ by 0.001 N, since the mean normal speed was nearly zero. The corresponding physical reaction on the robot is

$$F_{n,\mathrm{contact}} = n_s^\top \cdot f_{\mathrm{contact}} = -F_{n,\mathrm{est}} = +78.451\,\mathrm{N}.$$

Moment comparison about t_1

The centre of compliance is at the TCP. It therefore introduces no virtual force-to-moment coupling:

$$m_{\mathrm{CoC}} = r_c \times f = 0.$$

The physical contact quantities shown in Figure 1 are

$$r_{\mathrm{contact/TCP}} = p_{\mathrm{contact}} - p_{\mathrm{TCP}},$$
$${}^{O}m_{\mathrm{contact,TCP}} = r_{\mathrm{contact/TCP}} \times {}^{O}f_{\mathrm{contact}},$$
$$M_{t_1,\mathrm{contact}} = t_1^\top \cdot {}^{O}m_{\mathrm{contact,TCP}}.$$

The grey interval is treated as quasi-static. The measured orientation error gives

$$\begin{aligned} e_{R,t_1} &= 0.126923\,\mathrm{rad} = 7.272^\circ, \\ M_{t_1,\mathrm{cmd}} &= K_{R,t_1} \cdot e_{R,t_1} \\ &= 5 \cdot 0.126923 \\ &= +0.635\,\mathrm{N\,m}. \end{aligned}$$

Moment equilibrium then requires the opposing surface-on-robot moment

$$\begin{aligned} M_{t_1,\mathrm{contact}} + M_{t_1,\mathrm{cmd}} &= 0, \\ M_{t_1,\mathrm{contact}} &= -0.635\,\mathrm{N\,m}. \end{aligned}$$

With the model-estimated force magnitude, this corresponds to the quasi-static equivalent lever

$$r_{\perp,\mathrm{qs}} = \frac{|M_{t_1,\mathrm{contact}}|}{|F_{n,\mathrm{est}}|} = \frac{0.635}{78.451} = 0.00809\,\mathrm{m} = 8.09\,\mathrm{mm}.$$

This is an equilibrium equivalent, not a measured contact location.

The stationary model estimate is $M_{t_1,\mathrm{est}} = +0.604\,\mathrm{N\,m}$, compared with $M_{t_1,\mathrm{cmd}} = +0.635\,\mathrm{N\,m}$. The difference is $0.031\,\mathrm{N\,m}$, or 4.87%.

Orientation change

The measured component about t_1 fell from 9.26° at contact entry to 1.99° at the endpoint. Thus,

$$\Delta\theta_{t_1} = 1.99^\circ - 9.26^\circ = -7.27^\circ.$$

The off-centre contact rotates the tool towards the plane. The positive rotational command develops in the opposing direction and retains the entry orientation.

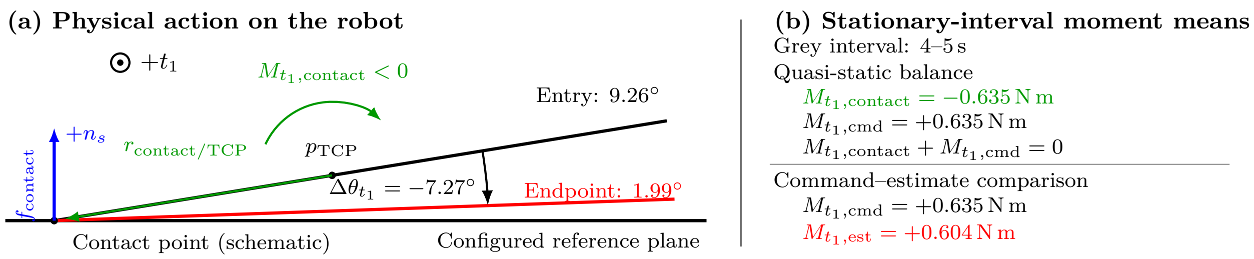


Figure 1: Contact response and endpoint moment comparison.

Normal-force summary. The quasi-static, commanded and model-estimated endpoint values are −80.313, −80.312 and −78.451 N. The estimate differs from the command by 2.32%.

Moment summary. The quasi-static, commanded and model-estimated endpoint values are +0.635, +0.635 and +0.604 N m. The estimate differs from the command by 4.87%. This single Contact Establishment trace is consistent in sign and magnitude for both components.

T-Mode Quasi-Static Consistency Test

The test uses the decoupled setup impedance in surface coordinates, with $r_c = 0$. The commanded components are

$$F_{n,\text{cmd}} = K_{p,n} \cdot e_n + D_{p,n} \cdot \dot{e}_n,$$
$$M_{t_1,\text{cmd}} = K_{R,t_1} \cdot e_{R,t_1} + D_{R,t_1} \cdot \dot{e}_{R,t_1}.$$

During each stationary hold, $\dot{e}_n \simeq 0$ and $\dot{e}_{R,t_1} \simeq 0$. The quasi-static predictions therefore reduce to

$$F_{n,\text{qs}} = K_{p,n} \cdot e_n, \qquad M_{t_1,\text{qs}} = K_{R,t_1} \cdot e_{R,t_1}.$$

The trial uses manual damping. The gains in the surface basis $[t_1, t_2, n]$ are

$$K_p = [2000, 2000, 1000]\ \text{N/m}, \qquad D_p = [10, 10, 175]\ \text{N s/m},$$
$$K_R = [15, 5, 50]\ \text{N m/rad}, \qquad D_R = [10.01, 10.01, 10]\ \text{N m s/rad}.$$

The centre of compliance is at the TCP, $r_c = 0$. The accepted nominal checks were approximately 20 mm and 6°. At exactly 20 mm, the force prediction is

$$|F_{n,\text{qs}}| = 1000 \cdot 0.020 = 20.0\,\text{N},$$

and at exactly $6^\circ = 0.104720\,\text{rad}$, the moment prediction is

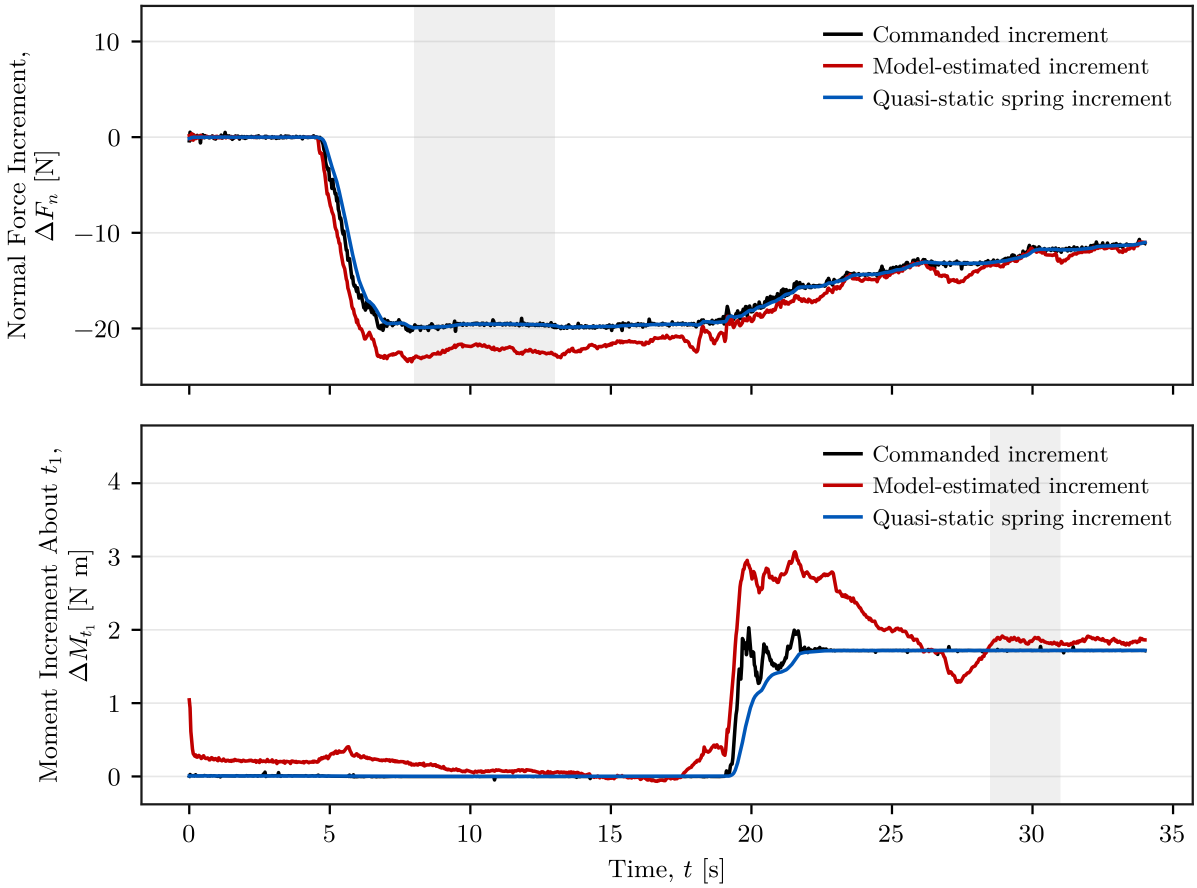
$$|M_{t_1,\text{qs}}| = 15 \cdot 0.104720 = 1.571\,\text{N m}.$$

T-Mode Consistency Results

Used impedance. The gains in the surface basis $[t_1, t_2, n]$ are

$$K_p = [2000, 2000, 1000]\,\mathrm{N/m}, \qquad K_R = [15, 5, 50]\,\mathrm{N\,m/rad},$$
$$D_p = [10.00, 10.00, 175.00]\,\mathrm{N\,s/m}, \qquad D_R = [10.010, 10.010, 10.000]\,\mathrm{N\,m\,s/rad}.$$

The centre of compliance is at the TCP, $r_c = 0$. The grey areas mark the stationary intervals: 8–13 s for F_n and 28.5–31 s for M_{t_1}. The measured means are calculated only from these intervals.



Component	Achieved	Quasi-static	Commanded	Estimated	Status
F_n	-19.65 mm	-19.650	-19.640	-22.292	consistent
M_{t_1}	6.565 deg	1.719	1.719	1.852	consistent

Normal force F_n.

$$F_{n,\mathrm{qs}} = K_{p,n} \cdot e_n = 1000 \cdot (-0.019650) = -19.650\,\mathrm{N}.$$

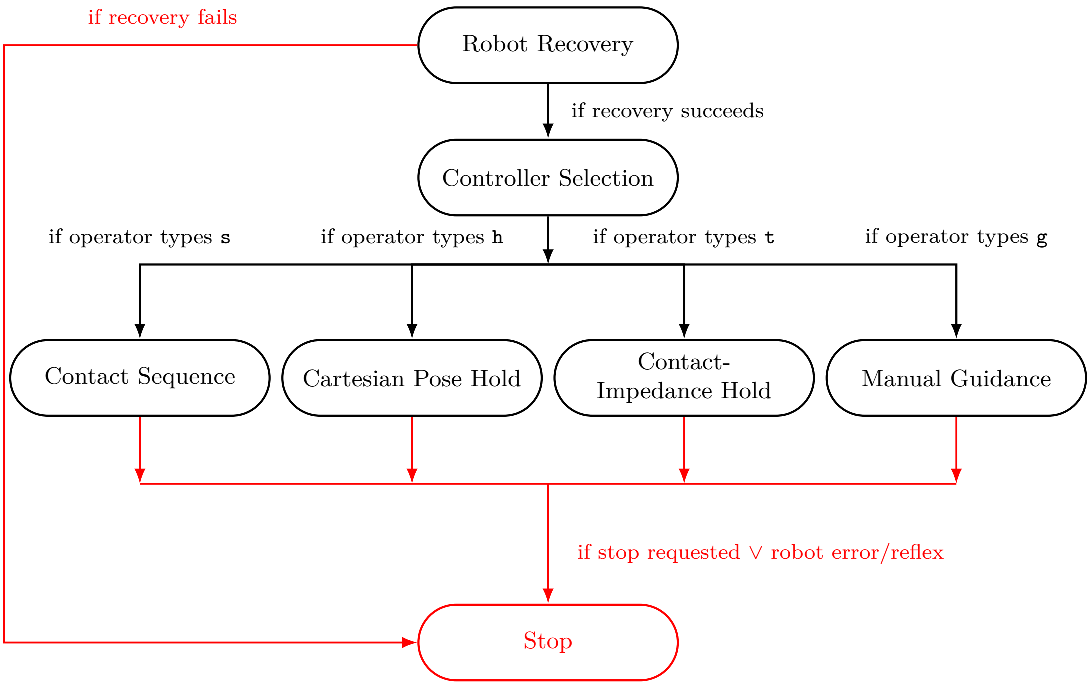
The commanded mean is -19.640 N. The quasi-static difference is 0.010 N (0.05%). The model-estimated mean is -22.292 N, a difference of 2.652 N (13.50%).

Moment M_{t_1}.

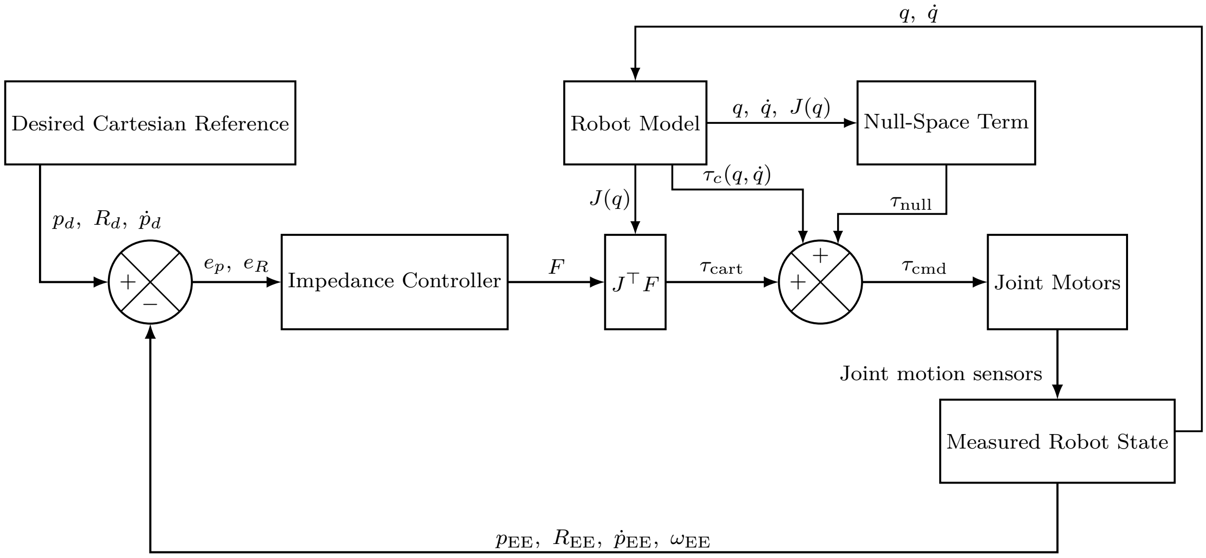
$$M_{t_1,\mathrm{qs}} = K_{R,t_1} \cdot e_{R,t_1} = 15 \cdot (6.565\pi/180) = 1.719\,\mathrm{N\,m}.$$

The commanded mean is 1.719 N m. The quasi-static difference is 0.000 N m (0.00%). The model-estimated mean is 1.852 N m, a difference of 0.133 N m (7.75%).

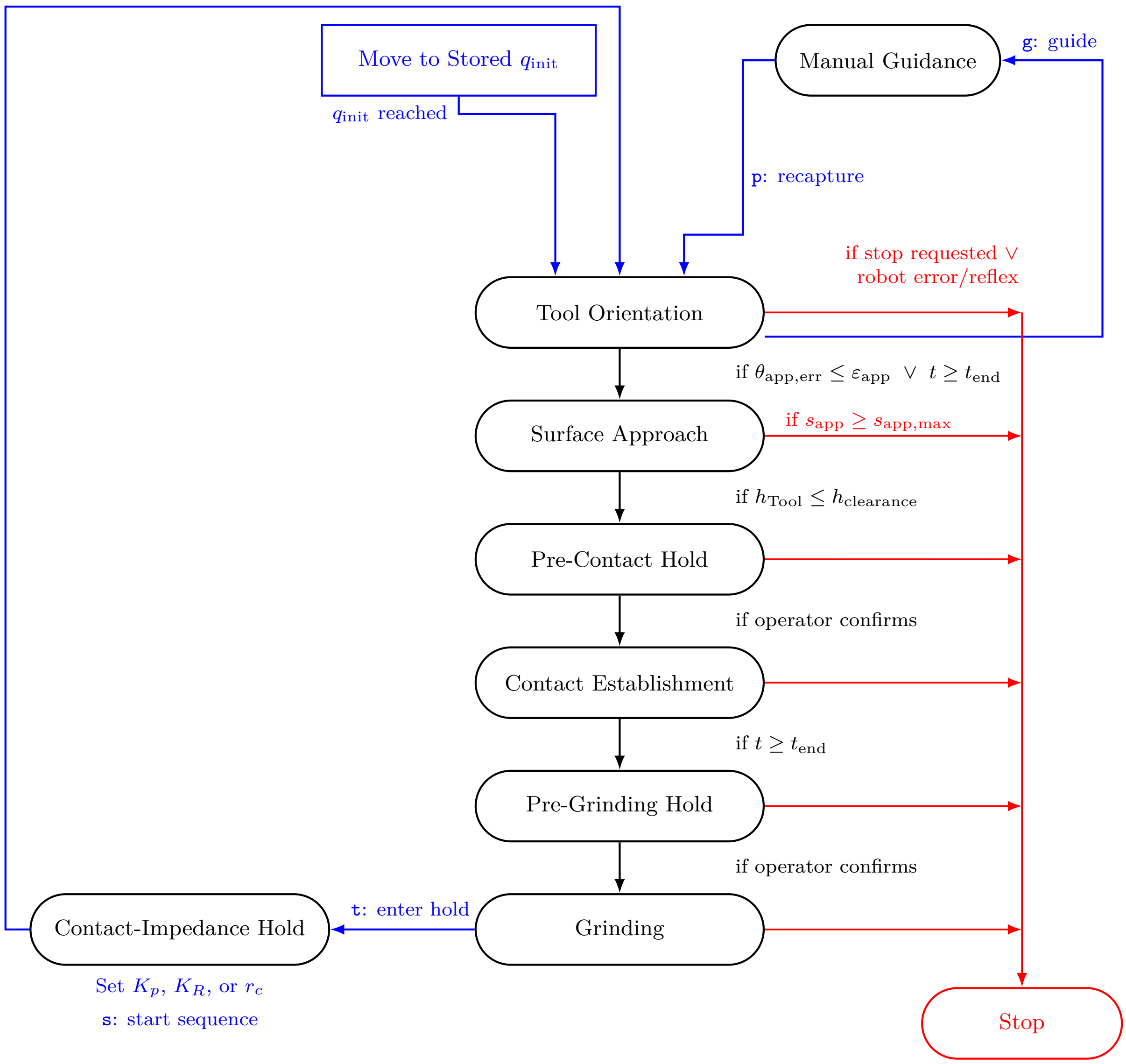
Controller State Flow



Cartesian Impedance Block Diagram



Surface-Contact Sequence



Cartesian Pose Hold